

Battery Storage Dispatch in Practice and in Theory: A Selective Literature Review of Modeling Approaches and Their Tradeoffs

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Abstract

Grid-scale battery energy storage systems (BESS) face a control problem at the intersection of stochastic markets, nonlinear degradation, and revenue stacking across timescales. Academic sizing tools (e.g. hydesign) and most published EMS formulations rely on deterministic mixed-integer LPs that assume perfect foresight over forecasted prices and resources. Commercial operators (Tesla Autobidder, Fluence Mosaic, Wärtsilä GEMS) instead use machine-learning-driven price forecasts coupled to stochastic optimization. This review surveys (i) what real BESS actually do for revenue, (ii) the dominant modeling-approach families (deterministic LP, stochastic programming, distributionally robust optimization, rolling-horizon MPC, reinforcement learning), and (iii) the major degradation-modeling choices (rainflow, semi-empirical, convex relaxation; post-hoc vs in-loop). Three takeaways relevant to the design of an open battery dispatch testbed: the perfect-foresight gap is real but modest ($\sim 11\text{--}16\%$) at hourly arbitrage, RL with in-loop nonlinear degradation can outperform MILP with linearized degradation by large margins on published benchmarks, and sub-hourly RL is degenerate without an exogenous fast-timescale driving signal.

1 Introduction

A grid-scale BESS dispatch policy decides, at each timestep, how much power to charge or discharge given forecasted prices, ancillary-service signals, renewable resource availability, grid constraints, and the cumulative degradation history of the battery. Three structural features make this problem distinctive:

1. **Multiple timescales.** Frequency-regulation signals update at seconds; real-time energy markets at 5–15 min; day-ahead at 1 h; replacement decisions at years. A single optimizer typically targets only one or two of these.
2. **Nonlinear, history-dependent degradation.** Rainflow cycle counting on the SoC trajectory is the standard offline accuracy metric; it is non-Markovian and lacks a closed-form representation suitable for direct use as an LP objective (19; 14).
3. **Stochastic exogenous inputs.** Day-ahead and intraday prices, wind, PV, and ancillary-service signals all carry forecast errors that are correlated, non-Gaussian, and regime-dependent.

The dominant academic approach—used in the canonical hydesign tool (11) and most published HPP-sizing studies—is to solve a deterministic MILP over a forecasted horizon, settle imbalance ex

post, then iterate post-hoc rainflow over the resulting SoC trace to determine battery replacements. Commercial operators do something materially different. This review surveys the gap.

2 What real BESS operators do

2.1 Revenue stacking has shifted to arbitrage

The CAISO market is the most-published example. As of end of 2024, 11.7 GW of installed BESS capacity earned an average \$51,000/MW-yr (down 35% from \$78,000/MW-yr in 2023) (2). Within that revenue, the share has shifted decisively: ancillary services (regulation + spinning reserve) dropped from \$13,279/MW-yr in 2023 to \$6,659/MW-yr in 2024 as the market saturated, while energy arbitrage rose from 3.7% of total revenue in 2022 to 24.5% by H1 2024. As of 2024, 43% of CAISO BESS were primarily dispatched for arbitrage. German indexes show 2024 revenues of ~100,000–220,000/MW-yr depending on portfolio and season (4).

Implication. Practitioner research on regulation-focused dispatch (e.g. heterogeneous-fleet RL for AGC tracking) is studying a shrinking revenue stream. Energy arbitrage, with its hourly-or-coarser timescale and direct exposure to forecast uncertainty, is now the central problem.

2.2 Commercial dispatch platforms are ML+stochastic, not deterministic LP

The three dominant commercial dispatch platforms—Tesla Autobidder (>100 GWh dispatched globally), Fluence Mosaic (16 GW deployed or awarded), and Wärtsilä GEMS (>9 GWh monitored)—all explicitly combine:

- Machine-learning-driven price/load/generation forecasts,
- Stochastic optimization with explicit uncertainty representation,
- Continuous online retraining and bid revision.

Tesla’s published description (16) lists “classical statistics, machine learning and numerical optimization” as constituent techniques. Fluence Mosaic uses “stochastic optimization models” built on ML price forecasts (6). Wärtsilä GEMS Pulse (released October 2024) is positioned explicitly around predictive analytics for revenue resilience under unpredictable market regimes (17).

Implication. The commercial state of practice is several model-classes ahead of deterministic LP. Academic tools that compare RL to deterministic LP and find the LP wins are not capturing the actual performance frontier.

3 Modeling approaches

3.1 Deterministic LP / MILP

The dominant academic baseline. Constraints are linear in SoC, ramp, and grid capacity; the objective is revenue minus a linear cycling-cost proxy. Hydesign (11) uses CPLEX with week-by-week batches over a 25-year plant lifetime. Variants include rolling-horizon MILP (re-solved at each step with updated forecasts; closer to practical operation but never explicitly stochastic).

Strengths. Convex, optimal under stated assumptions, well-tooled (CPLEX, Gurobi, Pyomo). Closed-form dual variables enable sensitivity analysis. Easy to extend with linear constraints.

Weaknesses. Cannot natively represent (i) nonlinear cycling costs, (ii) forecast distributions (only point forecasts), (iii) history-dependent state (e.g. rainflow’s running stack). Penalizing $|\Delta P|$ approximates degradation poorly when the cycle-depth-to-stress mapping is steep.

3.2 Stochastic programming

Two-stage and multi-stage variants represent uncertainty as a scenario tree: the first-stage decision (e.g. day-ahead bid) is followed by a second-stage decision (e.g. balancing-market settlement) conditioned on the realized scenario (9). Zhu et al. (20) apply this to hybrid wind-battery plants in spot and balancing markets.

Strengths. Captures forecast uncertainty by construction. Provides explicit risk-conditional value-of-information.

Weaknesses. Computational burden grows exponentially in scenarios; scenario reduction introduces approximation error. Distribution must be specified or sampled, often from a small empirical history.

3.3 Distributionally robust optimization (DRO)

Optimizes over the worst case within an ambiguity set of probability distributions, typically defined by a Wasserstein ball around the empirical distribution (18; 10). Avoids overfitting to a particular scenario sample.

Strengths. No need to specify the true distribution. Tractable reformulations exist for many problem classes. Tunable conservatism (ball radius).

Weaknesses. Conservatism can be hard to interpret economically. Ball-radius selection is itself a hyperparameter without a principled default.

3.4 Rolling-horizon model-predictive control (MPC)

Re-solves a (deterministic or stochastic) optimization at each step using the most recent forecast, applies only the first action, then re-measures. Now the dominant control framework in industry (? Evaluation 2025).

Strengths. Adapts to forecast updates. Can be combined with any objective (LP, MILP, stochastic). Empirically reduces cost vs rule-based by 3.9–47% depending on horizon and forecast quality (Evaluation 2025).

Weaknesses. Long horizons accumulate forecast errors; short horizons miss strategic opportunities (e.g. saving SoC for a known peak in 6 hours). Battery-aging models inside MPC must be approximate to remain solvable in real time.

3.5 Reinforcement learning (RL)

Frames dispatch as an MDP with state (SoC, prices, residuals, forecasts) and action (charge/discharge), then learns a policy via DQN, PPO, SAC, or similar (3; 13; 8). The MIT Cao et al. benchmark (3) reports a NN-DDQN policy beating MILP by 58.5% on lifetime profit when an accurate degradation model is included in the reward.

Strengths. (a) Reward can be arbitrary nonlinear (rainflow proxy, true cycling cost). (b) Policy is learned from sampled (forecast, realized) pairs so distributional robustness is implicit. (c) Inference is one forward pass—no per-step optimization. (d) Naturally handles continuous and high-dimensional state.

Weaknesses. (a) No optimality guarantee. (b) Sample-inefficient; PPO/SAC training is slower than MILP solve in many settings. (c) Degenerate when the underlying problem is convex with clean lookahead—collapses to a smooth approximation of greedy or LP (15). (d) Hyperparameter brittle; results often poorly reproducible.

3.6 Hybrid approaches

Recent work explores hybrids: LP solutions used as expert demonstrations to bootstrap RL via behavior cloning (Behavior Cloning 2024), or RL adjusting LP setpoints at fast timescales (two-tier). These do not yet have a settled performance frontier.

4 Degradation modeling

4.1 Rainflow

Rainflow cycle counting (ASTM E1049, equivalent to the Downing-Socie 4-point algorithm) extracts equivalent constant-amplitude cycles from a SoC trajectory. Each cycle is mapped to capacity loss via a stress function $f(\delta)$; total degradation is the sum. Rainflow is widely accepted as the offline accuracy benchmark (19).

Tradeoff. Non-Markovian: extracting cycles requires the full SoC stack. Shi et al. (14) construct a convex cycle-based relaxation that admits LP embedding; the relaxation introduces $<2\%$ error vs offline rainflow on representative traces but is still less accurate than full rainflow.

4.2 Semi-empirical / equivalent-cycle

Maps capacity loss directly to throughput (charge transferred), weighted by cycle-depth-to-stress factors. Linear or piecewise-linear; trivially LP-compatible. Adopted in NREL’s HPP degradation studies (12). Misses the rainflow stack’s dependence on cycle ordering.

Tradeoff. Tractable but inaccurate when cycle depth varies sharply within the operating window.

4.3 Post-hoc vs in-loop

Most academic dispatch optimizers (hydesign, most stochastic-LP work) compute degradation *post-hoc* on the optimized SoC trace, then update SoH and iterate. The optimizer itself sees only a linear cycling proxy. Cao et al. (3)’s 58.5% improvement over MILP comes specifically from

including the rainflow-based degradation in the RL reward *in the loop*—a representation MILP cannot replicate without convex relaxation.

Implication. The strongest case for RL over LP in this domain is degradation-fidelity, not forecast handling. A direct head-to-head requires a benchmark where in-loop nonlinear degradation matters at the dispatch timescale.

5 Forecast uncertainty: how much does perfect foresight buy?

Burlando et al. (1) report that a 1 MW/1 MWh BESS in the German continuous intraday market under a state-of-art forecast achieves 146,237/yr, only 11% below perfect foresight. Hancock et al. (7) show that Kendall’s τ rank correlation of forecasts (rather than MAE) is the primary determinant of dispatch value: $\tau \in [0.85, 0.95]$ captures 97–100% of perfect-foresight revenue, while persistence forecasts ($\tau \approx 0.4$) capture only 33%. Krishnamurthy et al. (9) report 16% gap between limited (24 h day-ahead + 1 h real-time) foresight and perfect foresight.

Implication. Above a moderate forecast-quality threshold, the perfect-foresight gap is small (~ 10 –15%) for energy arbitrage. RL trained on (forecast, realized) pairs can in principle close this gap, but the upside ceiling is modest. The much larger reported gain—Cao et al.’s 58.5%—comes from *degradation modeling fidelity*, not forecast handling.

6 Design formulations: sizing-dispatch coupling

Two architectures dominate.

Nested. Outer surrogate-based optimizer (e.g. Bayesian / efficient global optimization) over capacity decisions; inner LP-style EMS over a representative year. Hydesign (11) is a canonical example: `Parallel_EGO` on the outside, `ems_cplex` on the inside, post-hoc battery_replacement loop iterates SoH. The decoupling buys tractability but assumes the inner LP’s policy is the policy that will actually be deployed—an assumption the commercial-platform survey above suggests is wrong.

Joint. Single MILP/MINLP combining sizing variables and dispatch variables over a representative horizon. Computationally heavy; rarely scales beyond toy cases.

Hybrid. Sizing problem solves over discounted lifetime cash flows where dispatch is replaced by a learned RL policy or a faster surrogate. Open research direction; few published cases.

7 Synthesis: where the gaps actually live

Combining the above, the *empirically large* gaps in current BESS dispatch modeling are:

1. **Degradation fidelity.** In-loop rainflow vs LP-friendly linear penalty. Cao 2020 reports 58% on a representative arbitrage problem.
2. **Replacement scheduling.** Most academic tools post-hoc iterate; no formulation jointly optimizes dispatch and replacement timing.
3. **Multi-market revenue stacking.** Commercial platforms stack arbitrage + ancillary + capacity; academic tools usually pick one. Each adds a discrete decision (when to switch from one revenue stream to another) that LP cannot represent without large MILPs.

The *empirically modest* gaps:

1. **Perfect-foresight uplift at hourly resolution** ($\sim 11\text{--}16\%$). RL training on stochastic forecasts can close this, but the absolute upside is bounded.
2. **Sub-hourly RL on regulation signals.** Without a heterogeneous fleet or a constraint MILP cannot represent, online RL collapses to a smooth greedy heuristic (15).

8 Implications for an open battery-dispatch testbed

For an open RL testbed (such as `battery_gym`) targeting publishable comparison against LP baselines, the most defensible thesis is:

At hourly resolution, RL with in-loop nonlinear degradation reward outperforms LP with linearized cycling penalty, on Pareto frontier of revenue vs lifetime degradation, by a margin large enough to motivate replacement of LP-based EMS in tools like hydesign.

This statement is testable with a single-week or single-year benchmark. Three concrete pilot questions follow:

1. Replicate Cao et al. (3)’s 58% number with the published algorithm + a current LP baseline (e.g. hydesign’s `ems_cplex`). Sanity-check the gap.
2. Score hydesign LP solutions under in-loop rainflow reward, then score a learned RL policy on the same trace. Identify the dispatch decisions where they disagree (e.g. deep cycle vs shallow cycle on a steep stress-function regime).
3. Sweep α on the joint reward $R = \text{revenue} - \alpha \cdot D$ to map the Pareto frontier; report the LP-dominated and RL-dominated regions.

Each is doable in 1–3 days on a laptop. A negative result—LP within noise of RL—is itself publishable as a refinement of the Cao 2020 finding.

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